

DBMS COURSE PROJECT

Movie Recommendation System – Synopsis

1. Title of the Project

Movie Recommendation and Watchlist System

2. Problem Statement

With the increasing number of movies available on streaming platforms, users often find it difficult to decide what to watch. Many people spend a lot of time browsing through large movie collections without finding suitable recommendations quickly. Traditional movie lists do not personalize suggestions based on user preferences.

A Movie Recommendation System can solve this problem by maintaining a structured database of movies, users, ratings, and watchlists. Using this data, the system can provide personalized movie recommendations, allow users to search movies easily, and manage their personal watchlists.

3. Objectives of the Project

- Design a relational database to store movies, users, ratings, and watchlists.
- Allow users to search movies and view detailed information.
- Provide personalized movie recommendations based on ratings.
- Enable users to create and manage their movie watchlists.

4. Scope of the Project

- **Features Included:** Movie browsing, search functionality, rating system, personalized watchlist, and recommendations.
- **Limitations:** The system provides basic recommendations and does not include advanced machine learning algorithms.
- **Target Users:** Movie enthusiasts and general users looking for movie suggestions.

5. System Requirements

Software Requirements

- Database: MySQL / PostgreSQL / SQLite
- Frontend: HTML, CSS, JavaScript (React optional)
- Backend: Node.js / Python Flask / Java

Hardware Requirements

- Computer with minimum 4 GB RAM
- Modern web browser (Chrome / Edge / Firefox)

- Any operating system (Windows / Linux / MacOS)

6. Existing System

- Users often rely on generic movie lists without personalization.
- Browsing through large collections takes significant time.
- No centralized system for maintaining personal watchlists.

7. Proposed System

- A centralized database storing movies, ratings, and user preferences.
- Search functionality to quickly find movies.
- Personal watchlists where users can save movies.
- Basic recommendation system based on user ratings.

8. Frontend Design

- User Login / Registration Page
- Home Page displaying recommended movies
- Movie Search Page
- Movie Details Page with rating option
- User Watchlist Page
- Dashboard for managing movies and ratings

9. Expected Outcome

The Movie Recommendation System will provide users with an easy and interactive platform to explore movies and receive personalized suggestions. The system will help users quickly find movies that match their interests and maintain a personal watchlist. By using a relational database, the system ensures efficient data storage, fast search functionality, and better organization of movie information.